ADS1015 and ADS1115 - converting raw reading to voltage

Moderators: adafruit_support_bill, adafruit

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wsnlabz

Posts: 5
Joined: Fri Feb 22, 2013 7:54 am

ADS1015 and ADS1115 - converting raw reading to voltage

by wsnlabz »
Fri Feb 22, 2013 8:40 am

Hi,

I'm an electronics newbie. I've used non-differential ADCs before but am new to differential ADCs and am trying to figure out how to convert the raw readings from the ADS1015 and ADS1115 to voltages. In particular, I'm used to using formulas like V_in = analogRead(A0) / 2^10 * Vdd, but it seems that in these 2 ADCs Vdd is irrelevant, only FS matters.

I've read up the forums, example code, and the datasheet and just need someone to verify that the stuff mentioned below are correct.

Let:
V_in,0 = Input voltage to A0, represented as 2's complement 16-bit
V_in,1 = Input voltage to A1, represented as 2's complement 16-bit
FS = full-scale range, in volts
x = (value of conversion register >> 4) for ADS1015, value of conversion register for ADS1115

For both ADS1015 and ADS1115:

PGA=1, FS = 4.096
PGA=2, FS = 2.048
PGA=4, FS = 1.024
PGA=8, FS = 0.512
PGA=16, FS = 0.256

For ADS1015, when taking a single-ended measurement:
Vin,0 = x / 2^11 * FS

For ADS1015, when taking a differential measurement and MUX[2:0] == 000:
Vin,0 - Vin,1 = x / 2^11 * FS

For ADS1115, when taking a single-ended measurement:
Vin,0 = x / 2^15 * FS

For ADS1115, when taking a differential measurement and MUX[2:0] == 000:
Vin,0 - Vin,1 = x / 2^15 * FS

Questions:
1) Assuming Vdd is within the limits (say 3.3V), the formulas are absolutely independent of Vdd and only depend on FS?
2) When taking a single-ended measurement, I can't measure voltages greater than FS (e.g. If FS = 2.048V, measuring anything from 2.049V and greater will always give 32752)?



adafruit_support_bill

Posts: 91140
Joined: Sat Feb 07, 2009 10:11 am

Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **adafruit_support_bill** »
Fri Feb 22, 2013 9:29 am

1) True. The voltage reference is internally generated.
2) True. Anything >= FS will read FS.



alaada

Posts: 1
Joined: Sun May 04,

Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **alaada** »
Thu May 08, 2014 5:06 am

Hi

I reply to this old post because I have one question about voltage

2014 5:01 pm

reference of the ADS1115 (I read the data sheet and some topic about the ADS1115, but I am not able to find a definitive answer).

Is the voltage reference for the ADC = FS. For example, if I set the gain to 16, is my voltage reference equals to 0.256V?

Thank you

Regards



adafruit_support_bill

Posts: 91140
Joined: Sat Feb 07, 2009 10:11 am

Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **adafruit_support_bill** »
Thu May 08, 2014 5:42 am

Effectively, yes. I believe that internally the actual reference voltage remains the same, but the signal is amplified.

el_godo

Posts: 2
Joined: Thu Feb 13, 2014 9:49 am

Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **el_godo** »
Fri Jan 15, 2016 7:02 am

Hi,
I wonder whether I can switch the PGA after initialization. I have a submersible light sensor to measure light underwater: I let it sink down to characterize light distribution within the water column. My amplifier stage gives me 0 to 5V. I would like to try to use 2/3 gain until the signal falls below 4V and then switch to gain 1 for better resolution. This is fairly easy to test.
My concern is on the way up. The example code is vague on Vsig > Vlimit. Is Vsig > 4.096V < VDD a danger to the chip or will it simple return FS? Or will I fry the chip when my sensor reaches 4.096V and I do not adjust the gain quickly enough.
Other than Maxim, TI does not easily provide information on the components, so I hope to get an answer here. Thanks very much in advance!
Seb



Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **adafruit_support_bill** »
Fri Jan 15, 2016 7:12 am

adafruit_support_bill

Posts: 91140
Joined: Sat Feb 07, 2009 10:11 am

The specs are rather vague on that. But I believe you are OK as long as the input is <= VDD. It will just return full-scale if the signal exceeds the range for the gain setting.

el_godo

Posts: 2
Joined: Thu Feb 13, 2014 9:49 am

Re: ADS1015 and ADS1115 - converting raw reading to voltage

by **el_godo** »
Fri Jan 15, 2016 10:20 am

Thanks Bill, I'll try. After understanding that I will only have 11 bit resolution in the end I will change to the ADS1115 anyway, so no harm in frying the 1015. I will post my results to close the topic.
Cheers
Seb

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